

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Isolation makes colonisation difficult: only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them • Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists • Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) • Endemics have small ranges and populations so are vulnerable to extinction	4
01.2	• The birds evolved without mammal predators so nest on the ground / in burrows with no defence • Rats eat eggs and chicks; numbers fell to very low levels (45 and 12 pairs) • After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor	3
01.3	• Introduced mosquitoes carry avian malaria to which endemic birds have no resistance; birds survive only above about 1500 m where it is too cold for malaria • Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range	2
01.4	• Control of developments and visitor numbers (biosecurity, quarantine of imports); protected areas; captive breeding of endemics	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	
02.1	• Coordinating global data on biodiversity conservation (Red List) • Increasing understanding of the importance of biodiversity • Deploying nature-based solutions to global challenges in climate, food and sustainable development	3

Question	Marking guidance	Marks
02.2	- Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ - Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered - A species is placed in the highest category for which it meets any one criterion, so Critically Endangered	3
02.3	- Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed - Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) - Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations	3
02.4	Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
03.1	<i>Any 4 from:</i> - Copying the structural / functional adaptations of living organisms in the design of manufactured items - Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use - Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) - Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) - Termite mound convection copied for natural ventilation of buildings; hollow strutted bird bones in lightweight roofs; gecko toe adhesion; burr hooks - Velcro; lotus leaf self-cleaning glass; spider silk for airbags / body armour; bat / dolphin echolocation - ultrasound scanners	4
03.2	- Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores; in controlled doses these affect human physiology - Examples: morphine / codeine from poppies; quinine from cinchona bark (malaria); aspirin from willow bark; taxol from yew (cancers); diosgenin from Mexican yam (contraceptives, cortisone) - Animal examples accepted: AZT from a marine sponge (HIV), shark liver lipids (tumours)	3
03.3	- Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals - Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships)	2
03.4	- Squid giant nerve cells - nerve function / heart disease, stroke; - marsupials (kangaroo, wallaby pouch) - embryo development; hippo skin secretion - sunscreen / burns; sea urchin embryos - teratogen testing; armadillo - leprosy; marine sponge proteins - transplant rejection	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	

Question	Marking guidance	Marks
04.1	<i>Any 2 from:</i> • Hydrological cycle • Carbon cycle • Nitrogen cycle • Atmospheric circulation • Thermohaline circulation • Photosynthesis	2
04.2	• Enzymes reduce the activation energy of reactions • So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures	2
04.3	<i>Any 1 from:</i> • Haber process (nitrogen fixation) • Incineration of wastes (decomposition) • Many manufacturing processes	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
05.1	• Primary: succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) • Secondary: succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) • Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation	3
05.2	• Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form • The stable community that develops under this repeated disturbance is a plagioclimax • Examples: chalk grassland / hay meadows maintained by grazing or mowing; lowland heath by burning; coppiced woodland; Machair grassland; Broads grazing marshes	3
05.3	• If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost • Rainforest and coral reefs are climax communities so conservation aims to minimise human impact	2
05.4	• Extreme environments (deserts, polar) are dominated by abiotic factors; few species tolerate them; populations fluctuate and food webs are simple so changes in one food species have large effects • Stable environments allow many species to evolve inter-species relationships; populations are regulated by biotic factors; complex food webs give stability (rainforests, reefs)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	
06.1	• Readings drift so they may not be accurate; calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions • pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly • If a meter cannot be adjusted, record the percentage error and correct the results	3

Question	Marking guidance	Marks
06.2	• Same height above the ground • No obstacles to air flow; impeller type facing into the wind • Axis of rotation horizontal (impeller) or vertical (cup); same time period / averaging	3
06.3	• Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling • Evaporation cools the wet bulb more when humidity is low; the temperature difference is compared with a table to find humidity	2
06.4	<i>Any 2 from:</i> • Test strips with reagent bands compared with a colour chart (semi-quantitative) • Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength • Ion-selective electrode measuring electrode potential (other ions can interfere)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
07.1	• Cycling of materials (like biogeochemical cycles) - recycling, re-use • Energy from renewable sources • Human activities should support ecosystems / conserve biodiversity • Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis) • Diverse systems are more resistant to change • Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling	6
07.2	• Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae • Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs	2
07.3	• Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them • A common approach gives a level playing field, coordinated monitoring and support for less successful countries	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	
08.1	• Mars has low mass so weak gravity and could not retain a dense atmosphere • Very low atmospheric pressure means water would boil / evaporate away even at low temperatures • Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen	3

Question	Marking guidance	Marks
08.2	• Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil • Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured • Too close (Venus) - too hot; too far (Mars) - too cold	3
08.3	• The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface • The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions)	2
08.4	• The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun • Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted; the Moon's gravity keeps the tilt stable and creates tides	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
09.1	• Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs	1
09.2	• Impact depends on total population x per capita impact • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates	3
09.3	• Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need	2
09.4	• Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
10.2	• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Marine productivity: light, nutrients, upwelling; where fisheries occur • Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25